POWER BI CAPSTONE PROJECT

SuperStore Sales Dashboard

Sales Analysis & 15-Day Sales Forecasting

Student Name	Amrit Raj
Roll Number	23052941
Program	Power BI
Institute	ExcelR
Submission Date	April 21, 2026

Power BI Desktop | DAX | Time Series Analysis | Sales Forecasting

1. Project Title

SuperStore Sales Dashboard — An end-to-end Power BI project built on the SuperStore Sales dataset (2019–2020) that delivers an interactive sales dashboard and a 15-day sales forecast using time series analysis.

2. Objective

To contribute to the success of a business by utilizing data analysis techniques, specifically focusing on **time series analysis**, to provide valuable insights and accurate **sales forecasting**.

3. Description

The objective is broken down into the following four detailed components:

#	Component	Details
1	**Dashboard Creation**	Identify the KPIs, design an intuitive and visually appealing dashboard
2	**Data Analysis**	Provide valuable insights to business entities regarding the effectiveness
3	**Sales Forecasting**	Leverage historic data and apply time series analysis to generate accurate
4	**Actionable Insights**	Extract insights and actionable information that can drive strategic decisions

4. Dataset Description

The dataset **SuperStore_Sales_Dataset.csv** contains 5,901 records of retail orders placed between January 2019 and December 2020 across 4 regions of the United States. It covers 3 product categories, 3 customer segments, 4 shipping modes, and 3 payment methods.

Column	Data Type	Description
Order ID	Text	Unique identifier for each order
Order Date	Date	Date the order was placed (Jan 2019 – Dec 2020)
Ship Date	Date	Date the order was shipped
Ship Mode	Text	Standard Class / First Class / Second Class / Same Day
Customer Name	Text	Full name of the customer
Segment	Text	Consumer / Corporate / Home Office
City / State	Text	Geographic location of the order
Region	Text	East / West / Central / South
Category	Text	Furniture / Technology / Office Supplies
Sub-Category	Text	Phones, Chairs, Binders, Bookcases, etc.
Sales	Decimal	Revenue generated from the order (USD)
Quantity	Integer	Number of units ordered
Profit	Decimal	Profit earned from the order (USD)
Payment Mode	Text	Online / Cards / COD

5. Tech Stack

Component	Technology Used
BI Tool	Microsoft Power BI Desktop (Free)
Data Source	CSV Flat File — SuperStore_Sales_Dataset.csv (5,901 rows)
Data Transformation	Power Query Editor (M Language)

Calculations	DAX — 7 Measures + 4 Calculated Columns
Forecasting	Power BI Built-in Forecast (Analytics Pane) — 15 Days
Data Model	Single flat table with calculated columns
Visuals Used	Cards, Donut Charts, Area Charts, Bar Charts, Map, Line + Forecast

6. DAX Measures & Calculated Columns

The following DAX calculated columns and measures were created to power all KPI cards, charts, and the 15-day forecast on the dashboard:

Type	Name	DAX Formula
Calc. Column	Order Month	FORMAT([Order Date], "MMMM")
Calc. Column	Order Month Num	MONTH([Order Date])
Calc. Column	Order Year	YEAR([Order Date])
Calc. Column	Ship Days	DATEDIFF([Order Date], [Ship Date], DAY)
Measure	Total Sales	SUM('SuperStore_Sales_Dataset'[Sales])
Measure	Total Profit	SUM('SuperStore_Sales_Dataset'[Profit])
Measure	Total Orders	DISTINCTCOUNT('SuperStore_Sales_Dataset'[Order ID])
Measure	Total Quantity	SUM('SuperStore_Sales_Dataset'[Quantity])
Measure	Avg Ship Days	AVERAGE('SuperStore_Sales_Dataset'[Ship Days])
Measure	Profit Margin	DIVIDE([Total Profit], [Total Sales], 0)
Measure	Avg Order Value	DIVIDE([Total Sales], [Total Orders], 0)

7. Dashboard Pages & Screenshots

Super Store Sales Forecast-15 Days Forecast
Sales forecast-15 days
Sum of Sales
10K
5K
0K
Jan 2019
Apr 2019
Jul 2019
Oct 2019
Jan 2020
Order Date
Sales forecast-15 days
Sum of Sales
10K
5K
0K
Sep 2020
Order Date
Sales by State
California 0.34M
New York 0.19M
Texas 0.12M
Washington 0.09M
Pennsylvania 0.08M
Ohio 0.07M
Illinois 0.06M
Florida 0.05M
Michigan 0.05M
North Caro... 0.04M

8. Key Insights from the Dashboard

- California is the highest revenue state with $0.34M, followed by New York ($0.19M) and Texas ($0.12M)
- West region leads with 33% of total sales; South contributes the least at only 16%
- Consumer segment drives 48% of total sales, followed by Corporate (33%) and Home Office (19%)
- Standard Class shipping is the most used mode, contributing $0.33M — 70% of total shipments
- Office Supplies is the top-selling category ($0.64M), followed by Technology ($0.47M)
- COD (Cash on Delivery) is the most preferred payment mode at 43% of all orders
- Phones and Chairs are the top sub-categories by revenue across all regions
- Sales show a clear seasonal peak in Q4 (October–December) across both 2019 and 2020
- The 15-day forecast predicts continued growth maintaining the established seasonal upward trend

9. Actionable Recommendations

- Increase marketing investment in South region to bridge the 17% sales gap with the West
- Focus inventory planning around Q4 peak season to avoid stockouts of Phones and Chairs
- Introduce loyalty programs for the Consumer segment since it drives nearly half of all revenue
- Review Same Day shipping profitability — it contributes only 2% of total sales
- Leverage the 15-day forecast to pre-position stock in California and New York warehouses

10. Future Improvements

- Extend forecast from 15 days to 30 or 60 days using advanced DAX time intelligence functions
- Add Row-Level Security (RLS) so each regional manager sees only their own region's data
- Connect to a live SQL database instead of a static CSV for real-time dashboard refresh
- Publish to Power BI Service and schedule automatic daily data refresh
- Add customer-level profitability analysis to identify top customers by lifetime value
- Build a mobile-optimized layout for executive access on smartphones and tablets

11. Learning Outcomes

Through this project, I incorporated data analysis techniques, specializing in **time series analysis**, to deliver valuable insights, accurate **sales forecasting**, and interactive **dashboard creation**, driving business success. This project demonstrates end-to-end proficiency in Power BI — from data ingestion and Power Query transformation, to DAX calculations, interactive multi-page visualizations, and built-in 15-day forecasting using the Analytics pane.

This project uses the SuperStore Sales dataset (2019–2020) to build a two-page interactive Power BI dashboard featuring KPI tracking, regional sales analysis, segment and payment mode breakdown, and a 15-day sales forecast — matching industry-standard BI reporting practices.